Day 2 Report: RAG Base & Skeleton Pipeline

1. Title

Mental Health First-Response Chatbot – Day 2 Progress Report
Focus: Setup of RAG Pipeline and Knowledge Retrieval System

2. Objective

The objective of Day 2 was to build a basic Retrieval-Augmented Generation (RAG) pipeline that can retrieve relevant mental health support responses based on user input using a vector database.

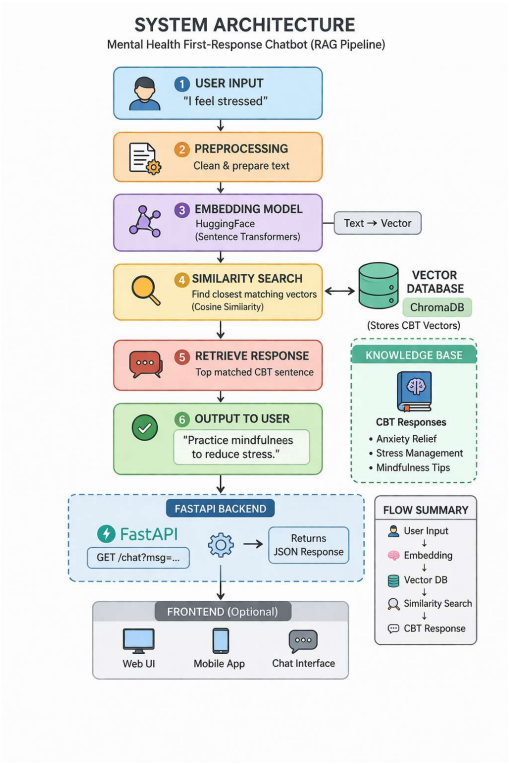
3. Overview of Work Done

On Day 2, we implemented a simple knowledge-based system using Cognitive Behavioral Therapy (CBT) techniques. A small dataset of CBT responses was created to act as the knowledge base. This data was then converted into vector form using embeddings and stored in a vector database (ChromaDB). The system retrieves the most relevant response based on user queries using similarity search.

4. Technologies Used

- **LangChain** – For building the AI pipeline
- **ChromaDB** – Vector database for storing embeddings
- **HuggingFace Embeddings** – To convert text into vectors
- **Sentence Transformers** – Underlying model for embeddings
- **FastAPI** – For backend API setup (basic integration)
- **Python (Google Colab)** – Development environment

5. System Architecture (Flow)



6. Implementation Details

- Created a **CBT-based knowledge dataset** containing mental health coping techniques.
- Used **HuggingFace embeddings** to convert textual data into vector representations.
- Stored vectors in **ChromaDB** for efficient similarity-based retrieval.
- Implemented **similarity search** to find the most relevant response for a given user query.
- Built a basic **FastAPI endpoint** to simulate backend communication.

7. Sample Output

Input:
"I feel stressed"

Output:
"Practice mindfulness to reduce stress."

8. Key Concepts Learned

- **CBT (Cognitive Behavioral Therapy):** Techniques to manage stress and anxiety
- **Embeddings:** Numerical representation of text capturing meaning
- **Vector Database:** Stores embeddings for similarity-based search
- **Similarity Search:** Finds closest matching response based on meaning
- **RAG:** Combines retrieval of knowledge with response generation

9. Achievements

- Successfully implemented a working RAG pipeline
- Verified retrieval of relevant CBT responses
- Built a basic backend structure using FastAPI
- Ensured the system runs without errors

10. Conclusion

The Day 2 objective of building a skeleton RAG pipeline was successfully achieved. The system is capable of retrieving meaningful mental health responses using vector similarity. This forms the foundation for further integration with NLP models and frontend interfaces in upcoming stages.

11. Future Work

- Integrate BERT-based sentiment analysis
- Expand CBT knowledge dataset
- Connect frontend chat interface
- Implement crisis escalation logic